Unit

6

The periodic table

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Periodic table, groups and periods
- Alkali metals, alkaline earth metals, halogens and noble gases

6.1 How to group elements together?

If you go to a supermarket, you will see that similar products are grouped together (Fig. 6.1). The number of elements is less than the number of products in a supermarket, but it is still useful to group similar elements together. How do chemists group elements together? What principles do they apply when doing that?



Fig. 6.1 In a supermarket, similar products are grouped together

Reference Website

- Refer to the following websites for information about various elements:
  - http://www.webelements.com (accessed July 2014)
  - http://www.rsc.org/periodic-table (accessed July 2014)
  - http://www.phy.ntnu.edu.tw/demolab/html.php?html=JavaScript/s_pertab (accessed July 2014)
- Refer to games on elements of the periodic table: http://www.sheppardsoftware.com/Elementsgames.htm (accessed July 2014)
- Refer to the Periodic Table Song: http://www.youtube.com/watch?v=zUDDiWtFtEM (accessed July 2014)

6.1

Inspecting samples of elements in the periodic table.

6.2 The periodic table

Chemists group elements with similar properties together. This gives rise to the **periodic table**. In the periodic table, all the elements are arranged in order of increasing atomic number. N1

N1 In the periodic table the elements are arranged in increasing order of atomic number, NOT neutron number or relative atomic mass.
✘ ✘

periodic table 週期表

A full periodic table can be found at the front of this book.

Fig. 6.2 is a simplified periodic table showing the first 36 elements (atomic number from 1 to 36).

N3 Refer to page T22.

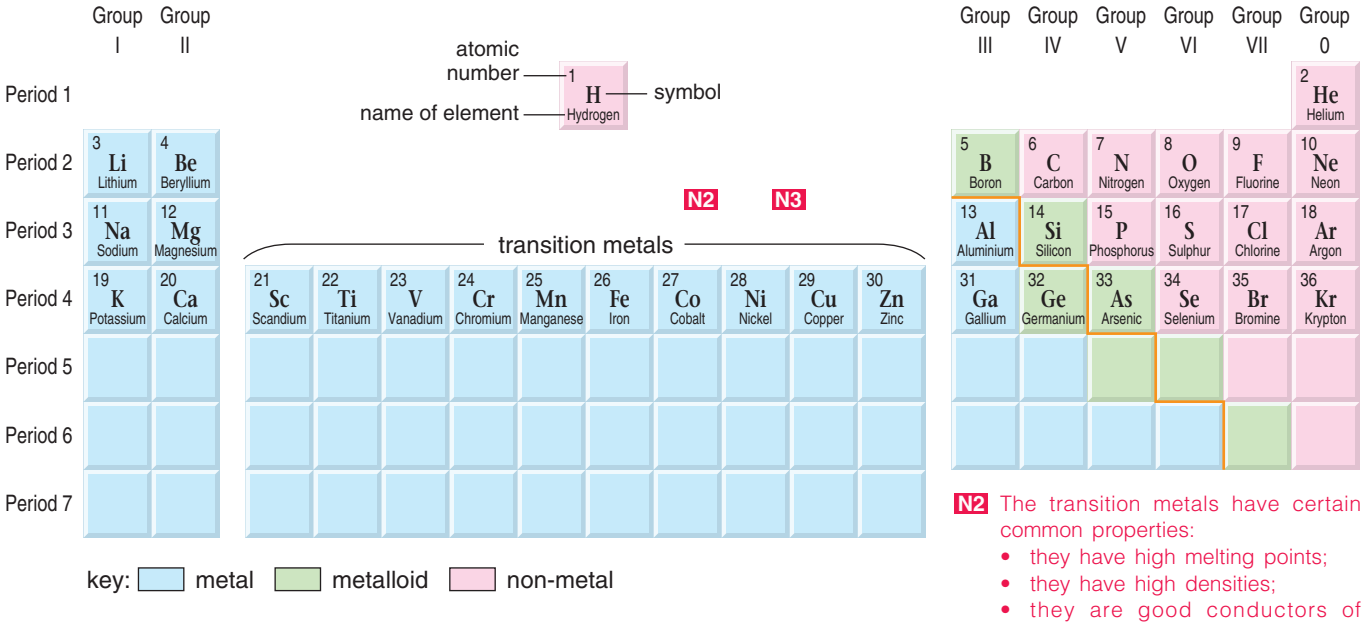


Fig. 6.2 A simplified periodic table showing the first 36 elements

N2 The transition metals have certain common properties:

- they have high melting points;
- they have high densities;
- they are good conductors of electricity;
- they are hard and rigid;
- they commonly form coloured compounds.

These elements are called transition metals because they have properties which are intermediate or transitional between elements on the left side of the table (Group I and II) and elements on the right side of the table (Group III to 0).

The horizontal rows in the periodic table are called **periods**. The vertical columns in the periodic table are called **groups**. Groups are numbered from I to VII, followed by Group 0 (or Group VIII). Elements between Group II and III of the periodic table are **transition metals**.

The elements in green separate metals (to the left) from non-metals (to the right). These elements, such as silicon and germanium, are metalloids.

6.3 States of elements

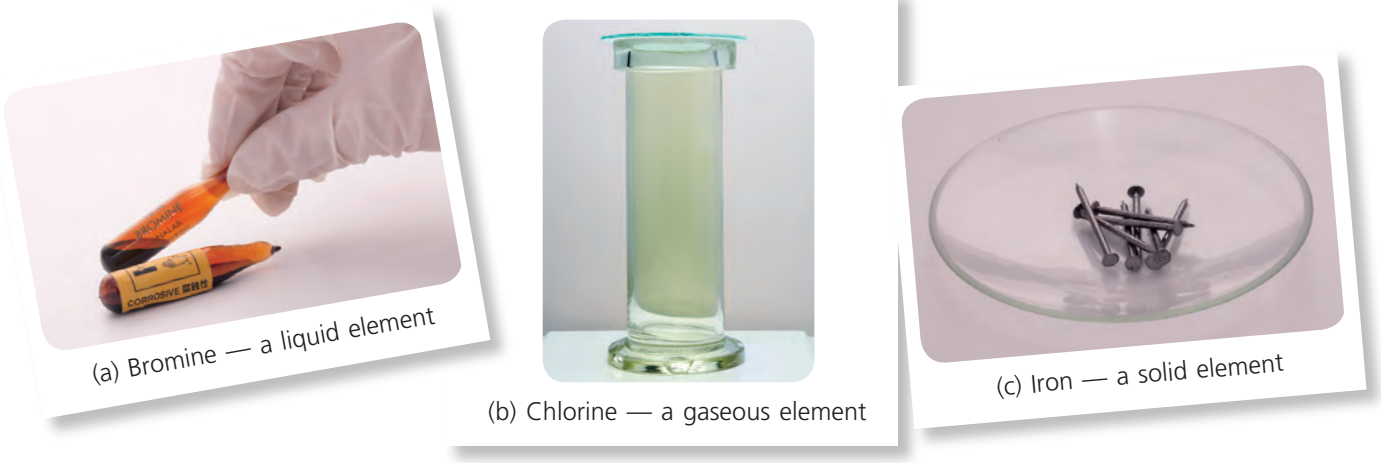
At room temperature and pressure means at 25 °C and 1 atmosphere.

Elements exist in different states at room temperature and pressure. Of the naturally occurring elements, at room temperature and pressure:

- two are liquids — mercury and bromine;
- eleven are gases — hydrogen, oxygen, nitrogen, fluorine, chlorine and six Group 0 elements (helium, neon, argon, krypton, xenon and radon);
- the rest are solids.

period 週期　　group 族　　transition metal 過渡金屬

Fig. 6.3a–c shows the states of some common elements.



(a) Bromine — a liquid element

(b) Chlorine — a gaseous element

(c) Iron — a solid element

Fig. 6.3 The states of some common elements at room temperature and pressure

Fig. 6.4 shows the states of the first 36 elements in the periodic table.

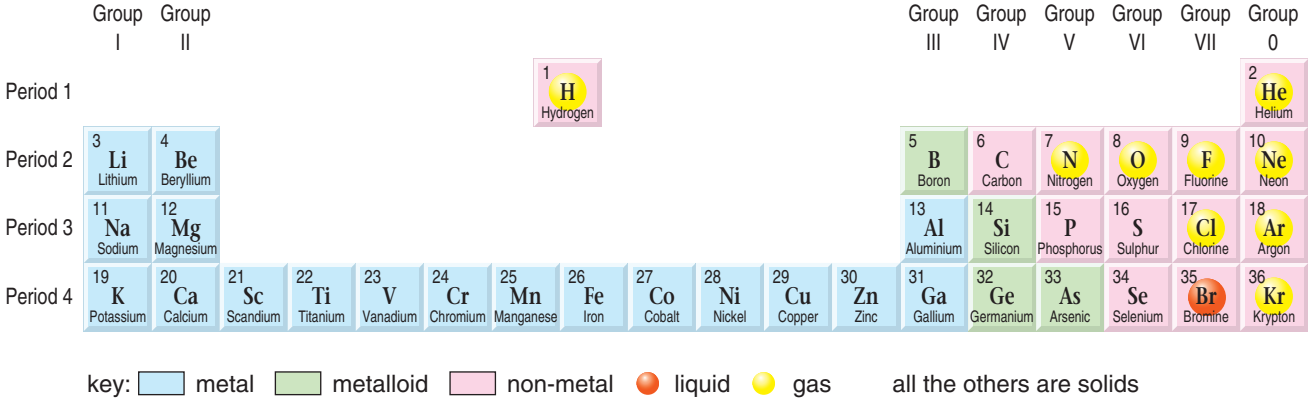


Fig. 6.4 The states of the first 36 elements in the periodic table

Practice 6.1

a) Fill in the boxes to show the state for each of the elements at room temperature (25 °C).

Element	Melting point (°C)	Boiling point (°C)	State at room temperature
W	–189	–186	gas
X	–110	–40	gas
Y	–7	58	liquid
Z	650	1 120	solid

b) Which of the elements, W, X, Y or Z, exists as a liquid at –70 °C? Explain your answer.

Element X.
X melts to give a liquid at –110 °C. Liquid X boils at –40 °C.
Thus, X is a liquid at –70 °C.

6.4 Position in the periodic table and electronic arrangement

Fig. 6.5 shows the electronic arrangements of atoms of the first 20 elements in the periodic table.

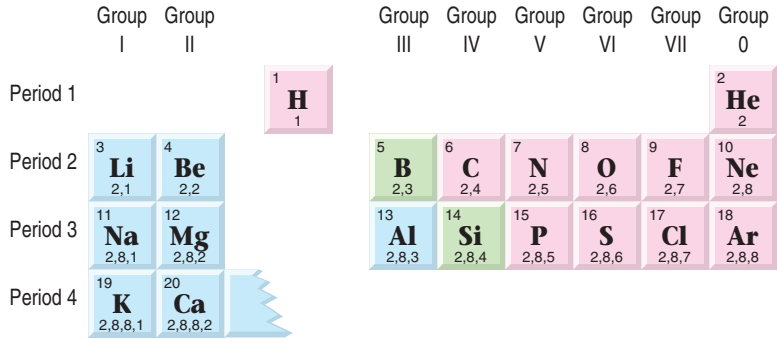


Fig. 6.5 The electronic arrangements of atoms of the first 20 elements in the periodic table

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Study Fig. 6.5. Suggest a common feature in the electronic arrangements of atoms of elements

i) in the same group; Atoms of elements in the same group have the same number of outermost shell electrons. The number equals to the group number of the element.

ii) in the same period.
Atoms of elements in the same period have the same number of occupied electron shells. The number equals to the period number of the element.

Groups — the vertical columns in the periodic table

A helium atom has 2 electrons.

The group number of an element is equal to the number of outermost shell electrons in its atom. For example, every member of Group II has 2 outermost shell electrons in its atom. Every member of Group 0 (or Group VIII) (except helium) has 8 outermost shell electrons in its atom.

✓ Group number of an element = number of outermost shell electrons in an atom of the element

✓ The chemical properties of an element depend on the number of outermost shell electrons in its atom.

Elements in the same group have similar chemical properties because they have the same number of outermost shell electrons in their atoms. These are the electrons which normally get involved when the elements bond to other things. However, there is a gradual change of **reactivity** of the elements as we move down the group.

Periods — the horizontal rows in the periodic table

CE 2008 Paper 2 Q22

Atoms of elements in the same period have the same number of occupied electron shells. The atom of each element in Period 1 has one occupied electron shell. The atom of each element in Period 2 has two occupied electron shells.

✓ Period number of an element = number of occupied electron shells in an atom of the element

6.5 Patterns across the periodic table

N4 The metallic character of elements decreases across a period while the non-metallic character increases.

Across a period, the elements change from metals through metalloids to non-metals (Fig. 6.6). N4

CE 2008 Paper 2 Q22

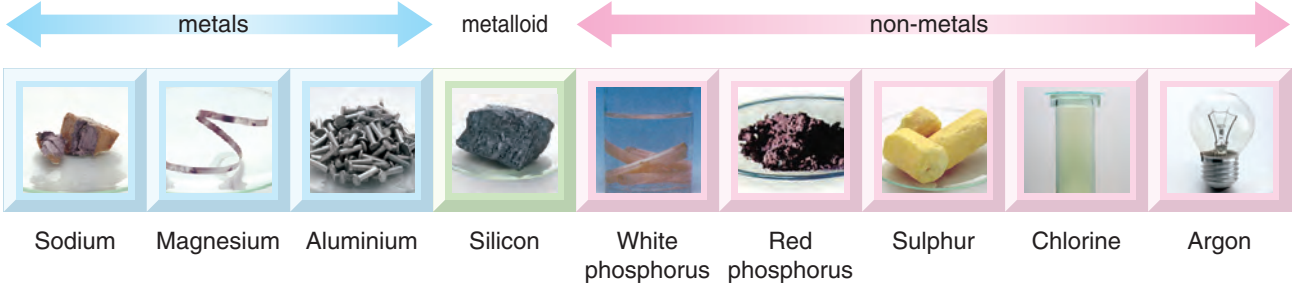


Fig. 6.6 Samples of the elements in Period 3

reactivity 活潑性

N5 Refer to page T22.

Table 6.1 shows some properties of the elements in Period 3. Notice that the reactivity of the elements also changes across a period. Apart from the noble gases, the most reactive elements are near the edges of the periodic table and the least reactive ones are in the centre. N5

Table 6.1

Some properties of the elements in Period 3

Element	Sodium	Magnesium	Aluminium	Silicon	Phosphorus	Sulphur	Chlorine	Argon
Type of element	← metals →			metalloid	← non-metals →			
State at room temperature and pressure	solid						gas	
Melting point (°C)	98	650	660	1 410	44	113	–101	–189
Boiling point (°C)	890	1 120	2 450	2 680	280	445	–34	–186
Electrical conductivity	good			moderate	poor			
Reactivity	reactive		moderately reactive	very unreactive	moderately reactive		very reactive	extremely unreactive

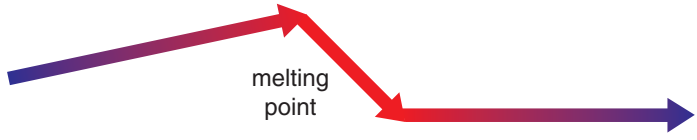
N6

Across Period 3, the atomic size of the elements shows a gradual decrease.

N6 Examination questions often ask about trends when going across a period of the periodic table.
e.g. going across the third period of the periodic table (from sodium to chlorine),
- the elements show a gradual change from having metallic property to having non-metallic property;
- the elements have the same number of occupied electron shells in their atoms;
- the atomic radius of elements decreases;
- the melting points and boiling points of elements do NOT vary gradually across a period.

Notice the following trends in the melting points of elements in Periods 2 and 3:

Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar



You should be able to explain these trends after learning the structures of the elements in Unit 9.

Example 6.1

Q The electronic arrangements of the atoms of three elements are shown in the table below:

Element	Electronic arrangement of atom
X	2,7
Y	2,8,5
Z	2,8,*p*,*q*

a) Elements X and Z belong to the same group of the periodic table.

i) To which group of the periodic table do they belong? Explain your answer.

ii) The atomic number of element Z is 35. What are the values of p and q?

b) To which period of the periodic table does element Y belong? Explain your answer.

A a) i) Group VII

Element X has 7 outermost shell electrons in its atom.

ii) $p = 18$

$q = 7$ N7

b) Period 3

Element Y has 3 occupied electron shells in its atom.

N7 Examination questions may ask students to predict the electronic arrangement of the atom of an element based on its atomic number.
e.g. the electronic arrangement of an atom of bromine (atomic number 35) is 2,8,*p*,*q*. What are the values of *p* and *q*?

Practice 6.2

The following is a part of the periodic table.

		H				He
Li	Be		B	C	N	O	F	Ne
Na	Mg		Al	Si	P	S	Cl	Ar
K	Ca							

a) Across a period, the elements show a gradual change in some physical properties. Suggest ONE such property.
Any one of the following:
Across a period, the elements show a gradual change (increase) in non-metallic character.
Across a period, the elements show a gradual change (decrease) in atomic size.

b) In the following parts, choose your answers from the symbols shown on the above periodic table.

i) Write the symbol of a metal with two electrons in the outermost shell of its atom.
Be / Mg / Ca

ii) Write the symbol of an element with eight electrons in the outermost shell of its atom.
Ne / Ar

iii) Write the symbol of an element with four occupied electron shells in its atom.
K / Ca

c) Rubidium (Rb) is below potassium in Group I of the periodic table. Its atomic number is 37.

Write the electronic arrangement of a rubidium atom. 2,8,18,8,1

Worksheet Assessment Form
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Historical development of the periodic table

Tasks

You are going to search and present information on the historical development of the periodic table.

1 Your teacher will divide the class into groups. Each group should give a 4-minute presentation. Support your presentation with various aids, such as timelines, diagrams, etc.

2 Write a short report of not more than 300 words to summarize your findings.

Hints for the search

1 Which chemist arranged elements into groups he called 'triads'? Why did he do that?

2 What is the 'Law of octaves'? Which chemist discovered the 'Law of octaves'?

3 What were the major findings of the Russian chemist Mendeléev?

Refer to page T26.

Reference websites

1 A summary of the historical development of the periodic table
http://www.ausetute.com.au/pthistor.html

2 A history of the development of the periodic table
http://www.bpc.edu/mathscience/chemistry/history_of_the_periodic_table.html

3 Website of the Royal Society of Chemistry
http://www.rsc.org/periodic-table/history
Click on 'Development of the Periodic Table' for the relevant articles.

Decision Making

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Grouping the elements

Imagine that a group of *astronauts* is trapped in another universe in which the elements are different from those found in our universe. A total of 13 elements have been identified.

Your teacher will give you a set of data cards of the 13 elements. Each card lists some properties of a particular element. N8 Refer to page T28.

1 Arrange these elements into an alternative periodic table. You may use the following steps:

a) Arrange the elements in order of increasing relative atomic mass.

b) Place the elements in a number of different groups. Each group should include elements with similar properties.

2 Do there appear any elements not yet discovered? If so, what are the likely properties of these elements?

3 A new element is discovered with a relative atomic mass of 25.8. Predict

a) where the element will fit in the alternative periodic table;

b) the likely properties of the element.

Refer to page T27.

astronaut 太空人

6.6 Group I elements — alkali metals

Francium is a radioactive element.

The six elements in Group I are lithium, sodium, potassium, rubidium, caesium and francium. These elements react with water to form alkalis. Hence they are called the **alkali metals** (Fig. 6.7). They are very reactive metals. Fig. 6.8a–b show samples of sodium and potassium respectively.

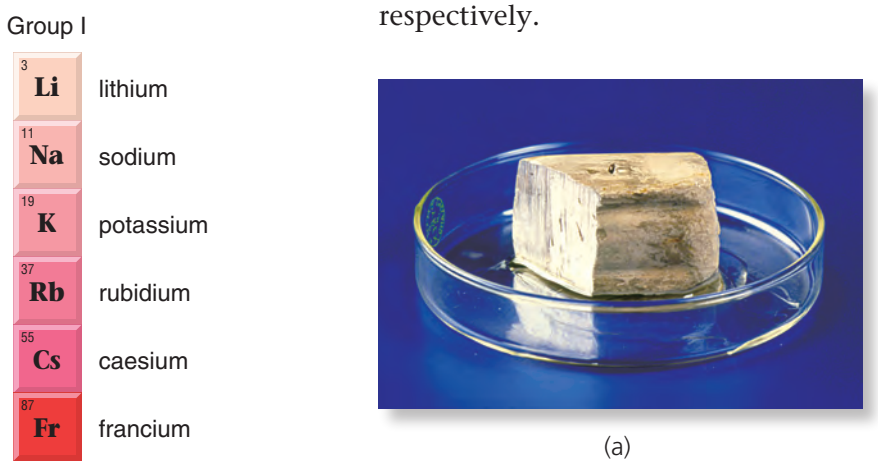
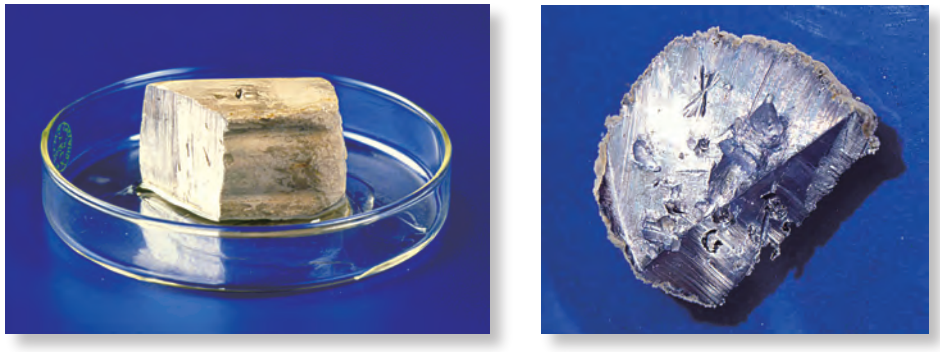


Fig. 6.7 The alkali metals in Group I



(a) (b)

Fig. 6.8 (a) Sodium (b) Potassium

№9 The number of occupied electron shells in an atom of successive Group I elements increases by one as we move down the group.

The number of occupied electron shells in an atom of successive Group I elements increases by one as we move down the group.

6.2
Investigating the properties of alkali metals.

Fig. 6.9 shows the electron diagrams and electronic arrangements № 9 of atoms of the first three Group I elements.

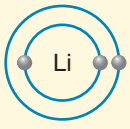
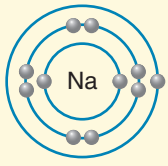
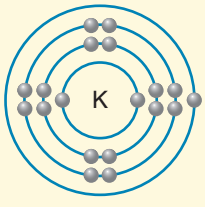
Element	Electron diagram of atom	Electronic arrangement of atom
Lithium		2,1
Sodium		2,8,1
Potassium		2,8,8,1

Fig. 6.9 The electron diagrams and electronic arrangements of atoms of the first three Group I elements

alkali metal 鹼金屬

Table 6.2 lists some physical properties of Group I elements. The melting and boiling points of the elements are very low for metals, and get lower as we move down the group (Fig. 6.10). N10

N10 The atomic radius of the elements increases down the group as electrons are added to successive electron shells and thus further away from the nucleus.

Table 6.2

Some physical properties of Group I elements

Element	State at room temperature and pressure	Melting point (°C)	Boiling point (°C)	Density (g cm^{-3})
Lithium	solid	180	1 330	0.53
Sodium		98	890	0.97
Potassium		64	760	0.86
Rubidium		39	686	1.53
Caesium		29	669	1.88
Francium		27	677	—

CE 2010 Paper 2 Q22

N11 The melting point, boiling point, density and hardness of Group I elements are all very much lower than the average values for metals.

N11

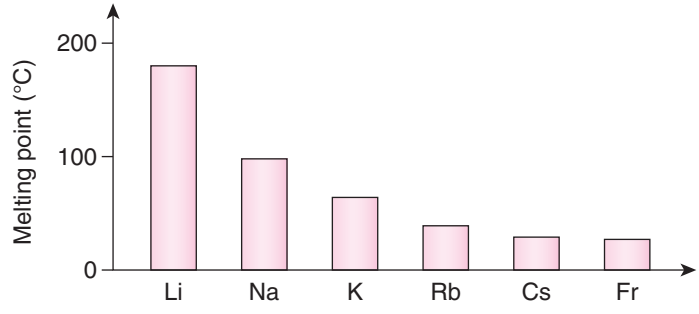


Fig. 6.10 Melting point pattern of Group I elements

CE 2005 Paper 2 Q36

Similarities of Group I elements

1 They are all soft and can be cut with a knife (Fig. 6.11). When freshly cut, they give a silver lustre but **tarnish** rapidly in air. N12

2 They all have low densities — lithium, sodium and potassium float on water.

3 Due to their high reactivity, they must be stored in paraffin oil to prevent them from reacting with air (Fig. 6.12). N13

N12 Among the Group I elements, lithium is the hardest.

The density of water is 1 g cm^{-3}. The densities of these metals are lower than that of water. Hence they float on water.

N13 Below are some safety precautions for handling Group I elements in the laboratory:
- wear safety glasses;
- use forceps;
- use a safety screen.

tarnish 失去光澤



Fig. 6.11 Sodium can be cut with a knife



Fig. 6.12 Sodium is stored in paraffin oil

Suppose a few drops of universal indicator solution are added to the solution formed in the reaction. The universal indicator solution would turn purple.

4 They all react vigorously with water to give hydrogen gas and an alkaline solution. For example, sodium reacts with water to give hydrogen and **sodium hydroxide** (the hydroxide makes the solution alkaline).

sodium + water ⟶ hydrogen + sodium hydroxide

5 They all react with certain non-metals to form compounds called **salts**. For example, sodium reacts with chlorine to form a salt called sodium chloride (Fig. 6.13a–b).

sodium + chlorine ⟶ sodium chloride



(a)



(b)

Fig. 6.13 (a) When sodium is heated and put in a jar of chlorine, it burns with a golden yellow flame (b) A white solid (sodium chloride) is formed

sodium hydroxide 氫氧化鈉 salt 鹽

AL 2010 Paper 1 Q9

Differences in reactivity of Group I elements

Group I elements are all very reactive. The reactivity of these elements increases as we move down the group (Fig. 6.14).

AL 2010 Paper 1 Q9

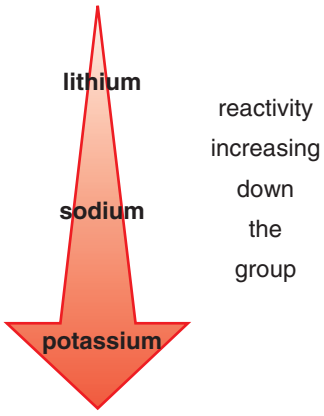


Fig. 6.14 Reactivity of Group I elements increases as we move down the group

The increasing reactivity trend from lithium to potassium is shown by their reactions with water (Table 6.3).

Table 6.3 Observations of reactions of the first three Group I elements with water

Element	Reaction with water
Lithium	• floats on water but does not melt • fizzes slowly
Sodium	• melts to form a silvery ball • fizzes quickly • moves rapidly on the water surface • sometimes the hydrogen ignites and burns with a golden yellow flame
Potassium	• melts to form a silvery ball • fizzes furiously • moves rapidly on the water surface • the hydrogen ignites and burns with a lilac flame

CE 2010 Paper 2 Q26

CE 2008 Paper 2 Q20

Sodium melts because its melting point is low and a lot of heat is produced by the reaction.

The lilac flame is due to contamination of the normally blue hydrogen flame by potassium compounds.

Fig. 6.15a–c show the reactions of lithium, sodium and potassium with water.



(a) Lithium reacts with water to give hydrogen steadily



(b) Sodium is more reactive than lithium; it moves rapidly on the surface of water



(c) Potassium reacts more vigorously with water than sodium does and catches fire immediately

Fig. 6.15 The reactions of the first three Group I elements with water

Video

Reaction of potassium with water

Predicting the chemical properties of unfamiliar elements

We can use the periodic table to predict chemical properties of unfamiliar elements. For example, sodium and potassium are Group I elements. Both of them react with water vigorously. Rubidium belongs to the same group. Thus, we expect it to react with water vigorously as well.

Example 6.2

Q Caesium is a Group I element below potassium in the periodic table.

a) How many outermost shell electron(s) is / are there in a caesium atom? Explain your answer.

b) Predict the state of caesium at room temperature and pressure.

c) Which of the metals, potassium or caesium, is more reactive? Explain your answer.

d) Write a word equation for the reaction between caesium and water.

e) Suggest ONE method to store caesium safely in the laboratory.

A

a) There is one outermost shell electron in a caesium atom. This is because caesium belongs to Group I of the periodic table.

b) Caesium is a solid at room temperature and pressure.

c) Caesium is more reactive than potassium.

The reactivity of Group I elements increases as we move down the group.

d) caesium + water ⟶ caesium hydroxide + hydrogen

e) It should be stored in paraffin oil.

Practice 6.3

A technician is making a film about the reactions of Group I metals with water. He drops a small piece of rubidium into a trough of water.

a) State TWO observations you expect.

Any two of the following:
- Melts to form a silvery ball.
- Fizzes furiously.
- Moves rapidly on the water surface.
- Catches fire.

b) The technician adds a few drops of universal indicator solution to the water in the trough after the reaction. The universal indicator solution turns purple. Name the reaction product that causes the universal indicator solution to turn purple.

Rubidium hydroxide

c) Suggest ONE safety precaution the technician should take when performing the reaction.

Wear safety glasses. / Use a safety screen.

6.7 Group II elements — alkaline earth metals

The six elements in Group II are beryllium, magnesium, calcium, strontium, barium and radium. These elements are found on the Earth and react with water to form alkalis. Hence they are called the **alkaline earth metals** (Fig. 6.16).

Fig. 6.17a–b show samples of magnesium and calcium respectively.

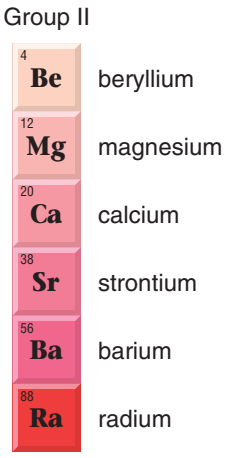
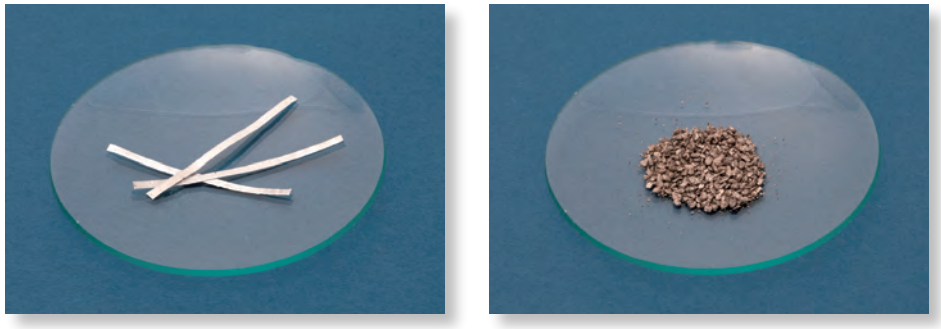


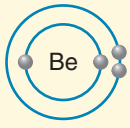
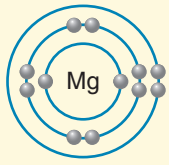
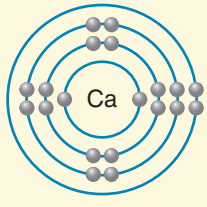
Fig. 6.16 The alkaline earth metals in Group II



(a) (b)

Fig. 6.17 (a) Magnesium (b) Calcium

Fig. 6.18 shows the electron diagrams and electronic arrangements of atoms of the first three Group II elements.

Element	Electron diagram of atom	Electronic arrangement of atom
Beryllium		2,2
Magnesium		2,8,2
Calcium		2,8,8,2

N14

Fig. 6.18 The electron diagrams and electronic arrangements of atoms of the first three Group II elements

N14 Examination questions may ask students to deduce the atomic number of an element from the known atomic number of another element in the same group.
e.g. X and Y are alkaline earth metals. The atomic number of X is *x*. Then the atomic number of Y could be *x* +18.

alkaline earth metal 鹼土金屬

N15 The hardness of Group II elements decreases down the group (beryllium is very hard while barium is very soft).

Table 6.4 lists some physical properties of the Group II elements N15 while Fig. 6.19 shows the melting point pattern of the Group II elements.

Table 6.4

Some physical properties of the Group II elements

Element	State at room temperature and pressure	Melting point (°C)	Boiling point (°C)	Density (g cm^{-3})
Beryllium	solid	1 280	2 480	1.85
Magnesium		650	1 120	1.74
Calcium		838	1 440	1.55
Strontium		769	1 384	2.60
Barium		725	1 640	3.51
Radium		700	1 137	—

CE 2010 Paper 2 Q22

N16 Examination questions often ask students to deduce the properties of unfamiliar elements based on familiar elements in the same group.
e.g. radium (Ra) and calcium (Ca) belong to the same group. Thus, it can be deduced that
- radium is a good conductor of electricity in the solid state.
- radium atom loses electrons readily to form a Ra^{2+} ion;
- radium tarnishes after exposed to air for some time;
- radium is more reactive than calcium.

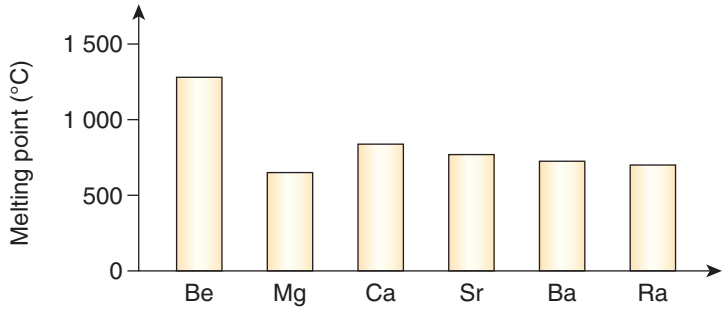


Fig. 6.19 Melting point pattern of Group II elements

Video
Grouping together metals with similar properties

Similarities of Group II elements N16

1 They all have low densities.

2 They are all reactive metals and react readily with dilute hydrochloric acid to give hydrogen gas (Fig. 6.20).

3 They all react with certain non-metals to form compounds called salts. For example, magnesium reacts with chlorine to form a salt called magnesium chloride.

magnesium + chlorine ⟶ magnesium chloride



Fig. 6.20 Magnesium reacts with dilute hydrochloric acid to give hydrogen gas

CE 2006 Paper 2 Q2
AL 2010 Paper 1 Q9

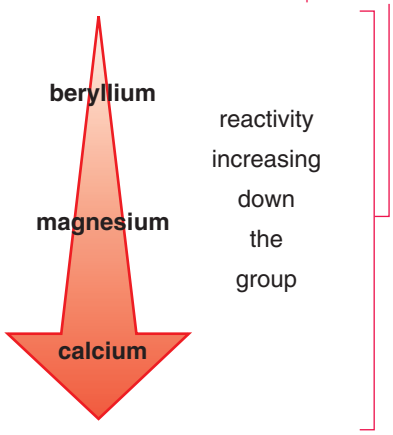


Fig. 6.21 Reactivity of Group II elements increases as we move down the group



Fig. 6.22 Calcium reacts with cold water to give hydrogen gas

Differences in reactivity of Group II elements

CE 2006 Paper 2 Q2
AL 2010 Paper 1 Q9

Group II elements are less reactive than Group I elements. The reactivity increases as we move down the group (Fig. 6.21).

The increasing reactivity trend from beryllium to calcium is shown by their reactions with water (Table 6.5). Among the three metals, only calcium reacts readily with water to give hydrogen gas (Fig. 6.22).

Table 6.5

Observations of reactions of some Group II elements with water

Element	Reaction with water
Beryllium	does not react with water
Magnesium	shows little reaction with cold water, but reacts readily with steam
Calcium	reacts readily, giving a steady stream of gas bubbles N17

Fig. 6.23 summarizes trends of some physical properties and N18 reactivity of Group I and II elements.

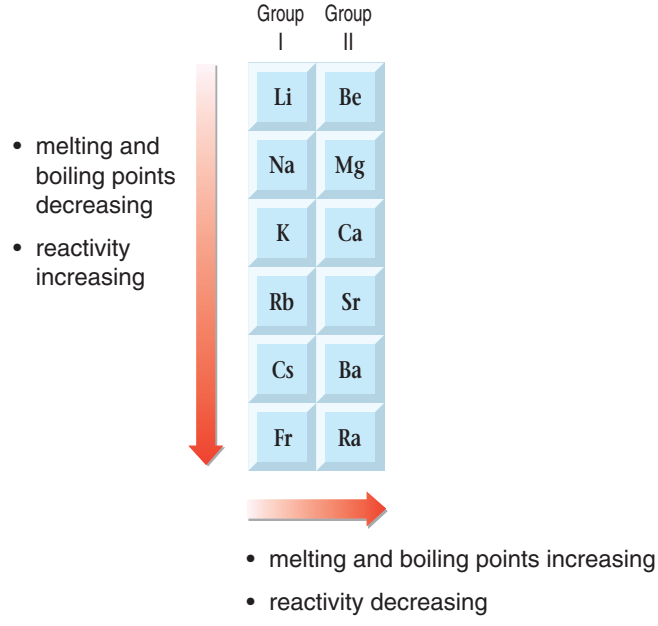


Fig. 6.23 A summary of trends of some physical properties and reactivity of Group I and II elements

N17 When barium is added to a trough of water containing phenolphthalein,
- barium sinks in the water;
- a colourless gas is liberated;
- an alkaline solution forms and the phenolphthalein turns pink.

N18 The following tables compare some properties of Group I and Group II elements in Periods 2 and 3.

Property \ Element	Sodium	Magnesium
Reaction with water	vigorous reaction	little reaction with cold water
Melting point (°C)	98	650
Boiling point (°C)	890	1 120
Hardness	magnesium is harder than sodium	

Property \ Element	Potassium	Calcium
Reaction with water	very vigorous reaction	steady reaction
Melting point (°C)	64	838
Boiling point (°C)	760	1 440
Hardness	calcium is harder than potassium	

Video
Determining which unknown element belongs to Group VII

6.3
Determining which unknown element belongs to Group VII.

We never have astatine in our school laboratory because it is a very unstable radioactive element.

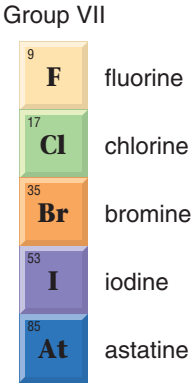


Fig. 6.24 The halogens in Group VII

6.8 Group VII elements — halogens

Group VII of the periodic table consists of the non-metals of fluorine, chlorine, bromine, iodine and astatine. These elements react with most metals to form salts. Hence they are called the **halogens** (which mean salt formers) (Fig. 6.24).

CE 2010 Paper 1 Q1(a)

Fig. 6.25 shows the electron diagrams and electronic arrangements of atoms of the first two Group VII elements.

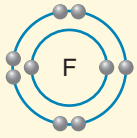
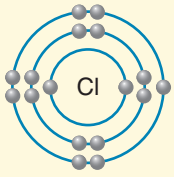
Element	Electron diagram of atom	Electronic arrangement of atom
Fluorine		2,7
Chlorine		2,8,7

Fig. 6.25 The electron diagrams and electronic arrangements of atoms of the first two Group VII elements

Table 6.6 lists some physical properties of the first four Group VII elements.

Iodine has a purple vapour.



Table 6.6
Some physical properties of the first four Group VII elements

Element	State at room temperature and pressure	Colour	Melting point (°C)	Boiling point (°C)	Density (g cm^{-3})
Fluorine	gas	pale yellow	–220	–190	0.00158
Chlorine	gas	greenish yellow	–101	–34	0.00296
Bromine	liquid	reddish brown	–7	58	3.11
Iodine	solid	black	113	184	4.93

N19

N19 The boiling points of chlorine, bromine and iodine increase down the group. Astatine is a solid at room temperature and pressure. It is soluble in heptane.

halogen 鹵素

N20 Examination questions often ask about the colours and states of halogens.
- Bromine is the only non-metal which is a liquid at room temperature and pressure.
- Iodine vapour is purple in colour.

There is a gradual change in state as we move down the group. Fluorine and chlorine are gases, bromine is a liquid, and iodine is a solid at room temperature and pressure (Fig. 6.26). There is also a gradual change in the intensity of colour, from pale yellow to black. N20

Fig. 6.27 shows the melting point pattern of the first four Group VII elements.

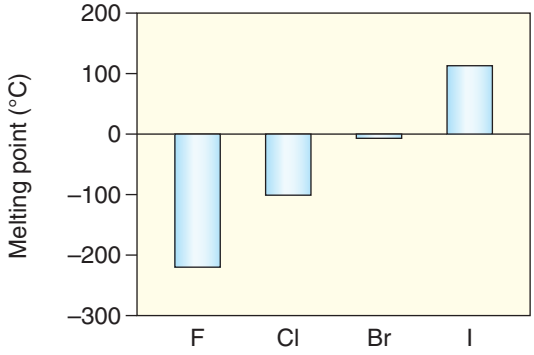


Fig. 6.27 Melting point pattern of the first four Group VII elements



Fig. 6.26 Chlorine, iodine and bromine (from left to right)

Do you know

Fluorine — the gas of Lucifer

Many early chemists were badly hurt or even killed as they tried to make pure fluorine. They called it 'the gas of Lucifer' (Lucifer actually means 'son of the morning'. Many recognize it as an alternative name for Satan, the fallen angel.) It was finally produced by the French chemist Henri Moissan, who died at the age of just 55. His life was almost certainly shortened by his work with fluorine.

Similarities of Group VII elements

1 They are all poisonous and smelly.

2 They are all non-metals.

3 They all react with metals to form compounds called salts. N21

N21 Chlorine, bromine and iodine all react with sodium sulphite solution.

Differences in reactivity of Group VII elements

CE 2007 Paper 2 Q22

The reactivity of these elements decreases as we move down the group (Fig. 6.28). N22

Group VII elements react with hydrogen to give hydrogen halides — hydrogen fluoride, hydrogen chloride, hydrogen bromide and hydrogen iodide. For example,

hydrogen + chlorine ⟶ hydrogen chloride

N22 The oxidizing power of chlorine is stronger than that of astatine.

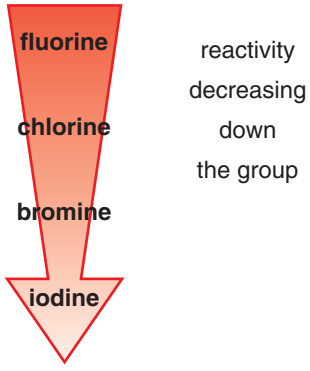


Fig. 6.28 Reactivity of Group VII elements decreases as we move down the group

The decreasing reactivity trend from fluorine to iodine is shown by their reactions with hydrogen (Table 6.7).

Table 6.7

Reaction conditions for halogens to react with hydrogen	
Group VII element	**Reaction with hydrogen**
Fluorine	reacts explosively in the dark
Chlorine	reacts explosively in sunlight or when heated
Bromine	reacts when heated
Iodine	partially reacts when heated

Practice 6.4

The following table lists some information about some halogens.

Element	Atomic number	Melting point (°C)	Boiling point (°C)
Chlorine	17	–101	–34
Bromine	35	–7	58
Iodine	53	113	184

a) Predict the melting point of fluorine.
Any value between –120 °C and –230 °C

b) What is the state of iodine at room temperature and pressure?
Solid

c) Describe how the reactivity of the halogens changes as the atomic number increases.
The reactivity decreases.

d) Astatine is another halogen. It is below iodine in Group VII.

A student makes some predictions about the properties of astatine. Which of the following predictions are likely to be true and which are likely to be false?

Use a tick '✔' to indicate your answer for each prediction.

Prediction	Likely to be true	Likely to be false
i) Astatine is a liquid at room temperature and pressure.		✔
ii) Astatine has a lower boiling point than iodine.		✔
iii) Astatine has 7 outermost shell electrons in its atom.	✔	
iv) Astatine is in Period 5 of the periodic table.		✔
v) Astatine reacts with iron more slowly than iodine does.	✔	

d) i) Astatine is a solid at room temperature and pressure.
ii) The boiling point of halogens increases as we move down the group. Thus, it is likely that astatine has a higher boiling point than iodine. The boiling point of astatine is 337 °C.
iii) Group VII elements have 7 outermost shell electrons in their atoms.
iv) Astatine is in Period 6 of the periodic table.
v) The reactivity of halogens decreases as we move down the group. Thus, it is likely that astatine reacts with iron more slowly than iodine does.

N23 All elements in Group 0 exist in gaseous state at room temperature and pressure.

There are no known compounds containing either helium or neon. Compounds of argon, krypton and xenon have been prepared, but with extreme difficulty.

6.9 Group 0 elements — noble gases

The six elements in Group 0 are helium, neon, argon, krypton, xenon and radon. They are called **noble gases** (Fig. 6.29) because they rarely react with other substances. N23

Fig. 6.30 shows the electron diagrams and electronic arrangements of atoms of the first three Group 0 elements.

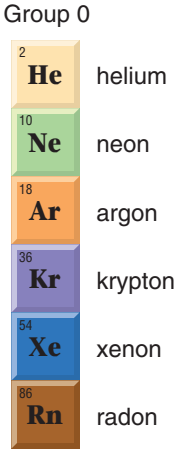


Fig. 6.29 The noble gases in Group 0

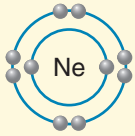
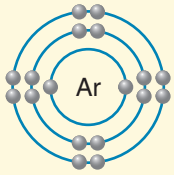
Element	Electron diagram of atom N24	Electronic arrangement of atom
Helium		2 (duplet structure)
Neon		2,8 (octet structure)
Argon		2,8,8 (octet structure)

Fig. 6.30 The electron diagrams and electronic arrangements of atoms of the first three Group 0 elements

N24 Remind students that NOT all atoms of noble gases have an octet structure in the outermost shell. Atoms of helium have a duplet structure.

Do you know

John Rayleigh and William Ramsay — the noble gases discovered

By the 1770s, chemists thought that the main components of the atmosphere had been well identified. But in 1892, John Rayleigh found that nitrogen isolated from the air had a density slightly higher than that of nitrogen prepared from nitrogen compounds. One dm^3 of pure nitrogen gas generated from a chemical reaction weighed 1.2505 g. On the other hand, one dm^3 of nitrogen gas generated from air by removing oxygen, carbon dioxide and water vapour weighed 1.2572 g (at the same temperature and pressure). A possible explanation was the presence of an unknown gas denser than nitrogen in the atmosphere.

Both William Ramsay and John Rayleigh tried to isolate this unknown gas using different methods. Each found evidence of the presence of an unknown gas in the atmosphere. This gas was a new element. Rayleigh and Ramsay announced their discovery in 1894, claiming that they had found a new element which did not fit into any group of the periodic table, and named it argon (from the Greek word 'argos', meaning 'inactive'). Ramsay went on to discover helium, neon, krypton and xenon.

noble gas 貴氣體

Table 6.8 lists some physical properties of the Group 0 elements. The melting point, boiling point and density of the elements increase as we move down the group.

Table 6.8

Some physical properties of the Group 0 elements

Element	State at room temperature and pressure	Melting point (°C)	Boiling point (°C)	Density (g cm^{-3})
Helium	gas	–270	–269	0.000179
Neon		–249	–246	0.000900
Argon		–189	–186	0.00178
Krypton		–157	–152	0.00373
Xenon		–112	–107	0.00589
Radon		–71	–62	0.00973

Fig. 6.31 shows the boiling point pattern of Group 0 elements.

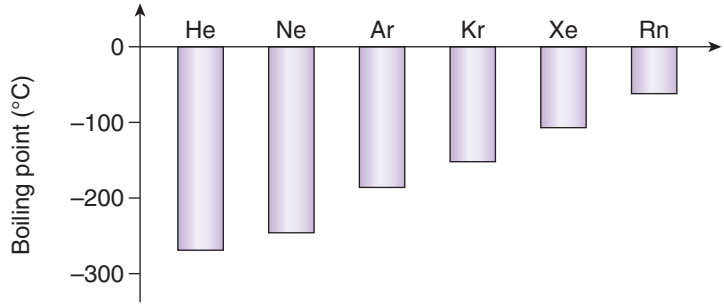


Fig. 6.31 Boiling point pattern of Group 0 elements

Similarities of Group 0 elements

1 They are all colourless gases at room temperature and pressure.

2 They all have very low melting and boiling points.

3 They are all very unreactive.

Stability of Group 0 elements

Each noble gas atom has 8 electrons in its outermost shell, except for helium, which has two (Fig. 6.30). An outermost shell of 8 electrons (or 2 electrons for helium) is extremely stable. It is this stability that makes the noble gases so unreactive.

The helium atom has only one electron shell which can hold only two electrons.

For other elements, atoms combine so that there are 8 electrons in the outermost shell (or 2 electrons for the lighter elements close to helium). This tendency to acquire the electronic arrangements of atoms of noble gases is referred to as the **octet rule**.

✓ When elements react, their atoms tend to do so in a way that results in an outermost shell containing 8 electrons (or 2 electrons for the lighter elements close to helium).

Uses of Group 0 elements

As Group 0 elements are unreactive, they are safe to use. Table 6.9 lists uses of the first three Group 0 elements.

DSE 2012 Paper 1B Q1(c)

Table 6.9

The uses of the first three Group 0 elements

Noble gas	Use(s)	Reason(s)
Helium	• in balloons and airships	• low density • unreactive
Neon	• in advertizing signs	• glows red when an electric current is passed through it
Argon	• filling electric light bulbs	• does not react with the metal filament in a light bulb

N25

CE 2008 Paper 1 Q12(d)

N25 It is NOT appropriate to fill an electric light bulb with air as the tungsten may react with air (oxygen) at high temperature.



Fig. 6.32 Helium is used in party balloons



Fig. 6.33 Neon is used in advertizing signs



Fig. 6.34 Argon is used to fill electric light bulbs

Game
Driving out aliens

octet rule 八隅體規則

6.10 From atoms to ions

Atoms can obtain the stable electronic arrangements of atoms of noble gases by gaining or losing electrons.

✓ A simple **ion** forms when an atom either loses or gains one or more electrons. It is either positively or negatively charged.

Positive ions — cations

Take calcium as an example. A calcium atom has an electronic arrangement of 2,8,8,2. It can obtain the stable electronic arrangement of an argon atom (2,8,8) by losing two electrons.

N26 Examination questions often ask students to describe the number of subatomic particles in an ion.
e.g. the doubly charged cation formed from $^{55}_{25}Mn$ contains 23 electrons and 30 neutrons.

When a calcium atom loses two electrons, a calcium ion forms (Fig. 6.35). The ion has 20 protons but 18 electrons only. Thus, the calcium ion is a doubly charged positive ion. We can represent it by the symbol Ca^{2+}. N26

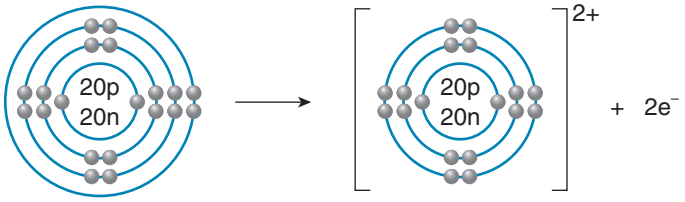


Fig. 6.35 A calcium atom loses 2 electrons to form a doubly charged positive ion

✓ When an atom of an element loses one or more electrons, it forms a positive ion. A positive ion is called a **cation**.

ion 離子 cation 陽離子

The name of a simple negative ion ends with the suffix '-ide'.

Notice that we write Ca^{2+} and N^{3-}, NOT Ca^{+2} and N^{-3}.
✔ ✔ ✘ ✘

Notice that we use 'dots' and 'crosses' to represent electrons of different atoms. These dots and crosses are just a means to count the electrons in different atoms. Actually electrons of all atoms are the same.

Negative ions — anions

Take nitrogen as an example. A nitrogen atom has an electronic arrangement of 2,5. It can obtain the stable electronic arrangement of a neon atom (2,8) by gaining three electrons.

When a nitrogen atom gains three electrons, a nitride ion forms (Fig. 6.36). The ion has 7 protons but 10 electrons. Thus, the nitride ion is a triply charged negative ion. We can represent it by the symbol N^{3-}.

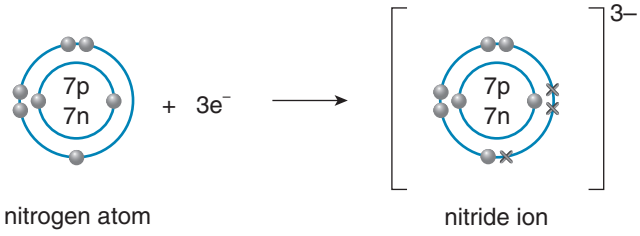


Fig. 6.36 A nitrogen atom gains 3 electrons to form a triply charged negative ion

✓ When an atom of an element gains one or more electrons, it forms a negative ion. A negative ion is called an **anion**. N27

N27 Examination questions often ask about the conversion of an atom of an element to form a cation / an anion. e.g. when an atom of a Group I metal is converted to a cation,
- an oxidation occurs;
- the cation has one less occupied electron shells compared with the atom;
- the cation and the atom have the same number of protons.

When an iodine atom is converted to an iodide ion,
- a reduction occurs;
- the iodide ion has the same number of occupied electron shells as the iodine atom;
- the iodide ion and the iodine atom have the same number of protons.

6.11 Predicting the charge on an ion

Table 6.10 lists some common ions with the stable electronic arrangements as those of atoms of noble gases.

Atoms of metals in Group I to III lose electrons so as to obtain the electronic arrangements of atoms of noble gases.

✓ Positive charge(s) on an ion formed from the atom of a metal = group number of the metal

Atoms of non-metals in Group V to VII gain '8 – group number' electrons so as to obtain the electronic arrangements of atoms of noble gases.

✓ Negative charge(s) on an ion formed from the atom of a non-metal = 8 – group number of the non-metal

anion 陰離子

Table 6.10

Some common ions with the electronic arrangements as those of atoms of noble gases

	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group 0
Period 2	Li^+ ◀ lithium ion	Be^{2+} ◀ beryllium ion			N^{3-} nitride ion	O^{2-} oxide ion	F^- fluoride ion	
Period 3	Na^+ sodium ion	Mg^{2+} magnesium ion	Al^{3+} aluminium ion			S^{2-} sulphide ion	Cl^- chloride ion	
Period 4	K^+ potassium ion	Ca^{2+} calcium ion N29					Br^- bromide ion	

N28

CE 2011 Paper 2 Q1

◀ Li^+ and Be^{2+} have an electronic arrangements of 2 (i.e. the electronic arrangement of a helium atom).

N28 N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} have the same number of electrons as a neon atom. S^{2-}, Cl^-, K^+ and Ca^{2+} have the same number of electrons as an argon atom.

N29 Examination questions may give the charge carried by a stable ion of an unknown element and ask the possible atomic number of the element.
e.g. an element forms a stable X^{2+} ion. The atomic number of X may be 20 or 38.

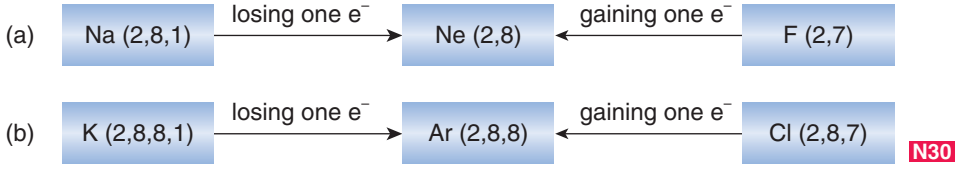
N30 Examination questions may describe the number of subatomic particles in the ion of an unknown element and ask students to deduce information about the element / its compound (e.g. the group or period to which the element belongs).
e.g. X^{2+} ion has an electronic arrangement of 2,8,8. Thus, an atom of X has an electronic arrangement of 2,8,8,2. X is calcium. The question may ask students to deduce information about the carbonate of X.

Group I elements have similar chemical properties because all their atoms obtain a stable electronic arrangement by losing one electron. For example,

- a sodium atom obtains the electronic arrangement of a neon atom by losing one electron (Fig. 6.37a);
- a potassium atom obtains the electronic arrangement of an argon atom by losing one electron (Fig. 6.37b).

Group VII elements have similar chemical properties because all their atoms obtain a stable electronic arrangement by gaining one electron. For example,

- a fluorine atom obtains the electronic arrangement of a neon atom by gaining one electron (Fig. 6.37a);
- a chlorine atom obtains the electronic arrangement of an argon atom by gaining one electron (Fig. 6.37b).



N30

Fig. 6.37 Atoms of Group I and VII elements obtaining a stable electronic arrangement

Practice 6.5

1 Complete the following table by giving the number of protons, neutrons and electrons in each ion.

Ion	Number of particles in the ion		
	protons	neutrons	electrons
$^{56}_{26}Fe^{2+}$	26	30	24
$^{69}_{31}Ga^{3+}$	31	38	28
$^{80}_{35}Br^{-}$	35	45	36

CE 2010 Paper 2 Q1

2 The electronic arrangements of three chemical species are shown below:

W : 2,8
X^{2+} : 2,8
Y^{-} : 2,8,8

X and Y are in the same period of the periodic table.
The electronic arrangement of an atom of X is 2,8,2 while that of Y is 2,8,7. Both atoms have 3 occupied electron shells.
Thus, X and Y are both Period 3 elements.

Which of the elements, W, X and Y, are in the same period of the periodic table? Explain your answer.

CE 2009 Paper 2 Q1

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Using argon to preserve leftover wine in an unfinished bottle

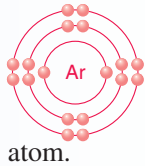
Ethanol is the main ingredient in wine. Oxygen in the air can oxidize ethanol in wine to form ethanal or ethanoic acid, giving the wine a sour taste. Argon is chemically unreactive and denser than air. Pumping argon into an unfinished bottle of wine displaces air from the bottle, thus preventing the wine from contact with air (Fig. 6.38). This means that one will never waste any wine because of oxidation.

Traditionally, nitrogen has been used in a similar manner for food preservation. However, argon is denser and fills spaces around the food more completely, making it a more efficient preservative agent.

The vacuum pump method is another wine preservation method. This is to pump air out of the unfinished bottle of wine and replace the cork. A vacuum is created inside the bottle. However, pumping air out may also remove volatile organic compounds that give the wine a pleasant odour.



Fig. 6.38 Argon is used to preserve leftover wine in an unfinished bottle



Questions

1 Draw an electron diagram of an argon atom.

2 Explain why argon is chemically unreactive.

3 Helium is also chemically unreactive. Suggest why helium is NOT used for preserving leftover wine in an unfinished bottle.

2 The outermost shell of an argon atom has a stable octet structure.
3 Helium is less dense than air. It cannot displace air from the bottle.

Have you mastered?

Key terms

periodic table 週期表 33 period 週期 34 group 族 34 transition metal 過渡金屬 34

reactivity 活潑性 37 alkali metal 鹼金屬 41 tarnish 失去光澤 42

sodium hydroxide 氫氧化鈉 43 salt 鹽 43 alkaline earth metal 鹼土金屬 46

halogen 鹵素 49 noble gas 貴氣體 52 octet rule 八隅體規則 54 ion 離子 55

cation 陽離子 55 anion 陰離子 56 nitride ion 氮離子 56 oxide ion 氧離子 57

sulphide ion 硫離子 57 fluoride ion 氟離子 57 chloride ion 氯離子 57

bromide ion 溴離子 57

Checklist

After studying this unit, you should be able to

- ☐ understand that elements in the periodic table are arranged in order of increasing atomic number;
- ☐ appreciate the periodic table as a systematic way to arrange elements;
- ☐ define the group number and period number of an element in the periodic table;
- ☐ relate the position of an element in the periodic table to its electronic structure and vice versa;
- ☐ relate the electronic arrangements to the chemical properties of Group I, II, VII and 0 elements;
- ☐ describe differences in reactivity of Group I, II and VII elements;
- ☐ predict the chemical properties of unfamiliar elements in a group of the periodic table;
- ☐ relate the stability of noble gases to the octet rule;
- ☐ describe the formation of simple ions from atoms;
- ☐ draw electron diagrams of cations and anions;
- ☐ predict the ions formed from metallic and non-metallic atoms by using information in the periodic table.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 In the periodic table, all the elements are arranged in order of increasing atomic number.

2 The vertical columns in the periodic table are called groups. Groups are numbered from I to VII, followed by Group 0 (or Group VIII).

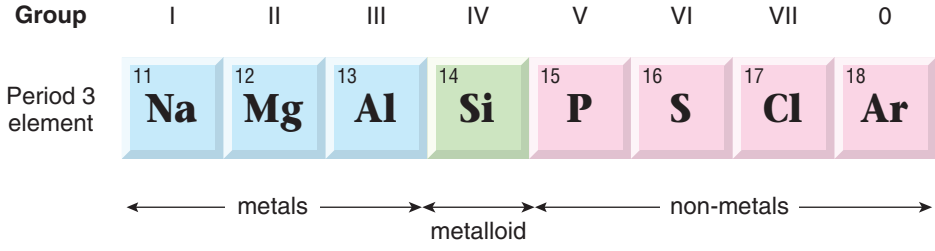
Group number of an element = number of outermost shell electrons in an atom of the element

3 The chemical properties of an element depend on the number of outermost shell electrons in its atom.

4 The horizontal rows in the periodic table are called periods.

Period number of an element = number of occupied electron shells in an atom of the element

5 Across a period in the periodic table, the elements change from metals through metalloids to non-metals.



6 Elements in the same group have similar chemical properties because they have the same number of outermost shell electrons in their atoms.

7 Group I elements — alkali metals

a) Their melting and boiling points are very low.

b) They are all soft and can be cut with a knife.

c) They all have low densities — lithium, sodium and potassium float on water.

d) They must be stored in paraffin oil to prevent them from reacting with air.

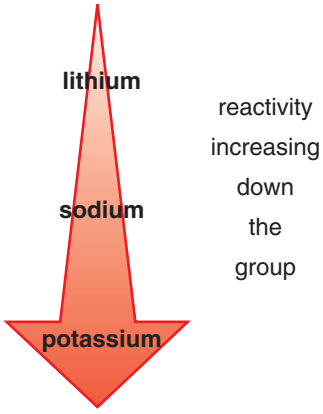
e) They all react vigorously with water to give hydrogen gas and an alkaline solution.

sodium + water ⟶ hydrogen + sodium hydroxide

f) They all react with certain non-metals to form compounds called salts.

Have you mastered?

g) The reactivity of these elements increases as we move down the group.



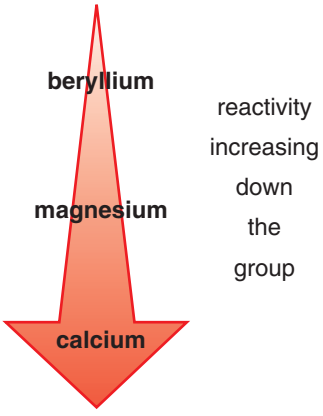
8 Group II elements — alkaline earth metals

a) They all have low densities.

b) They are all reactive metals and react readily with dilute hydrochloric acid to give hydrogen gas.

c) They all react with certain non-metals to form compounds called salts.

d) Group II elements are less reactive than Group I elements. The reactivity of these elements increases as we move down the group.



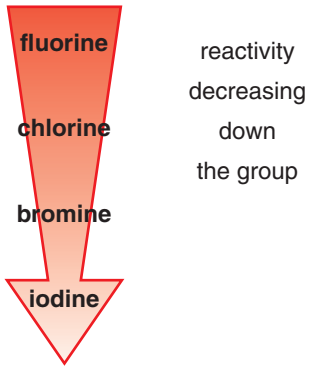
9 Group VII elements — halogens

a) They are all poisonous and smelly.

b) They are all non-metals.

c) They all react with metals to form compounds called salts.

d) The reactivity of these elements decreases as we move down the group.



10 Group 0 elements — noble gases

a) They are all colourless gases at room temperature and pressure.

b) They all have very low melting and boiling points.

c) They are all very unreactive.

11 When elements react, their atoms tend to do so in a way that results in an outermost shell containing 8 electrons (or 2 electrons for the lighter elements close to helium).

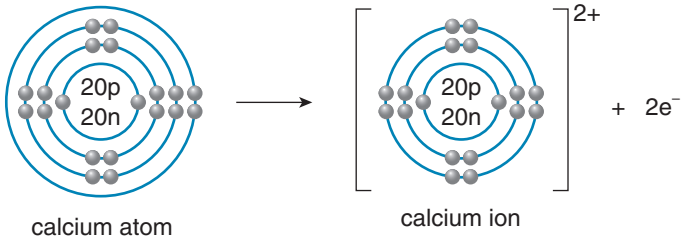
12 a) When an atom of an element loses one or more electrons, it forms a positive ion (i.e. a cation).

b) When an atom of an element gains one or more electrons, it forms a negative ion (i.e. an anion).

13 Atoms can obtain the stable electronic arrangements of atoms of noble gases by gaining or losing electrons.

14 a) Positive charge(s) on an ion formed from the atom of a metal
= group number of the metal

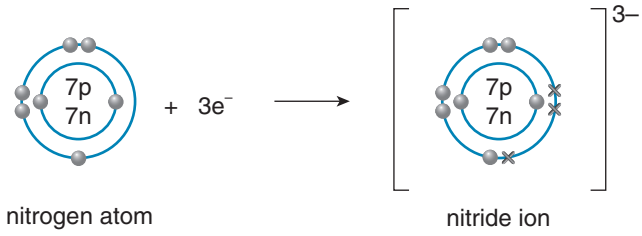
e.g. Calcium is a Group II metal. It forms a doubly charged positive ion (Ca^{2+}).



Have you mastered?

b) Negative charge(s) on an ion formed from the atom of a non-metal
= 8 – group number of the non-metal

e.g. Nitrogen is a Group V non-metal. It forms a triply charged negative ion (N^{3-}).



nitrogen atom nitride ion

	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group 0
Period 2	Li^+	Be^{2+}			N^{3-}	O^{2-}	F^-	
Period 3	Na^+	Mg^{2+}	Al^{3+}			S^{2-}	Cl^-	
Period 4	K^+	Ca^{2+}					Br^-	

Have you mastered?

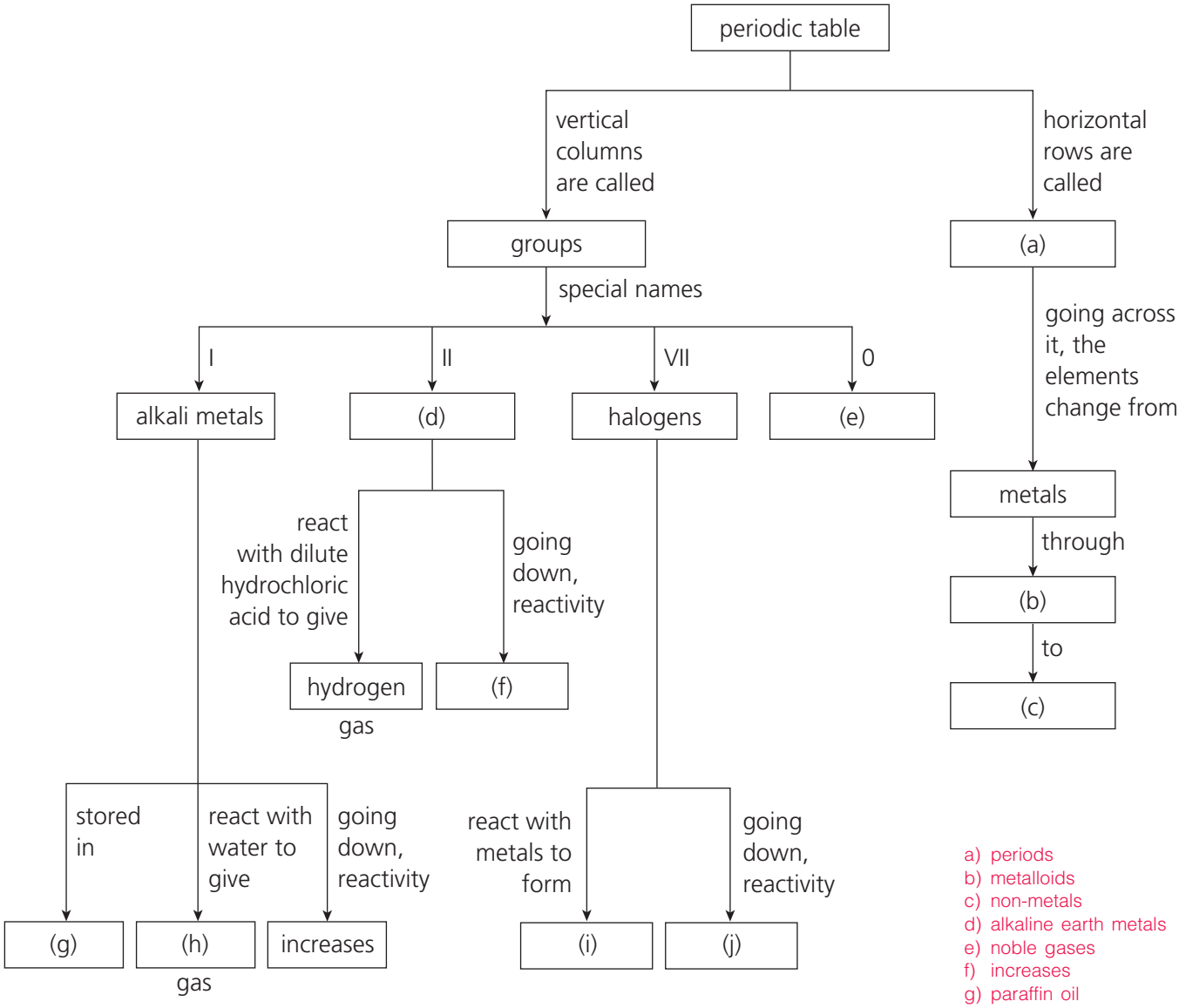
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the concept map using the following words:

alkaline earth metals, decreases, hydrogen, increases, noble gases, metalloids, non-metals, paraffin oil, periods, salts



a) periods
b) metalloids
c) non-metals
d) alkaline earth metals
e) noble gases
f) increases
g) paraffin oil
h) hydrogen
i) salts
j) decreases

2 The following questions are about the periodic table of the elements. Insert the missing words and / or numbers.

a) A horizontal row in the periodic table is known as a ___period___ .

b) All atoms of the elements in Group V have ___5 electrons___ in their outermost shells.

c) Elements in the same group have ___similar___ chemical properties.

d) The elements helium to radon are known as the ___noble___ gases.

(Edexcel GCE O Level, Paper 1, Jan. 2010, 3(a)–(c), (f))

3 This question is about elements in the periodic table.

Look at the list of elements.

aluminium, beryllium, calcium, fluorine, iron, lithium, neon, nitrogen, potassium, sulphur

Choose ONLY elements from the list to answer the questions.

Each element can be used once, more than once or not at all.

a) Write down the name of the element which gives a lilac flame when it burns.
Potassium

b) Write down the name of an element whose atoms have 7 electrons in its outermost shell.
Fluorine

c) Write down the name of the element with an electronic structure of 2,8,8,2 in its atom.
Calcium

(OCR GCSE Gateway Science, Chem. B, Unit 1, May 2011, 10)

4 Complete the table.

Particle	Number of protons	Number of neutrons	Number of electrons
$^{19}_{9}F$	9	10	9
$^{80}_{35}Br^-$	35	45	36
$^{39}_{19}K^+$	19	20	18

(Edexcel GCE O Level, Paper 1, May 2010, 4)

5 Complete the table.

A, B, C and D are NOT the symbols for the elements.

Element	Number of electrons in outermost shell	Formula of cation or anion usually formed
A	1	A^+
B	3	B^{3+}
C	6	C^{2-}
D	7	D^-

(Edexcel GCE O Level, Paper 1, Jan. 2010, 1(a))

6 The ***dominoes*** in the following figure carry the chemical symbols for metals and non-metals. They are formed by two squares and linked together to form a maze. Your task is to move from the '•' at the top left corner to the '•' at the bottom right corner. The rules are:

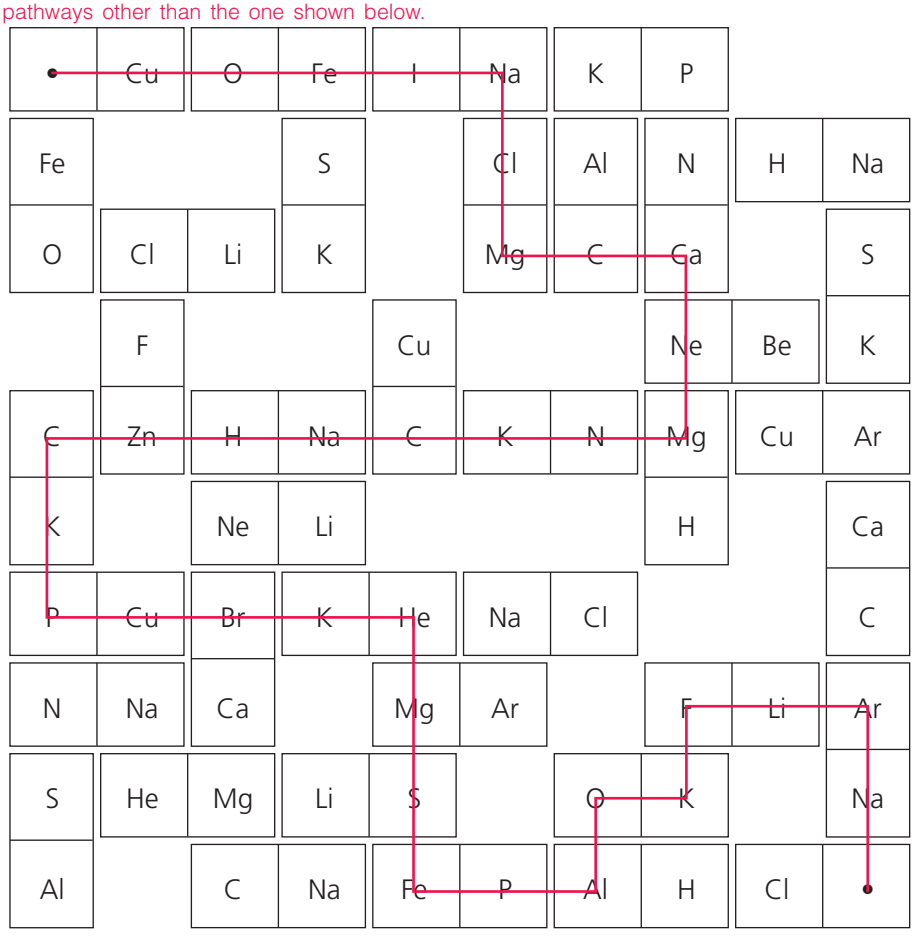
– Move to any square that are touching in each step

– No diagonal movement

– The pathway must follow the pattern: metal ⟶ non-metal ⟶ metal

Draw the pathway on the figure.

Students may use pathways other than the one shown below.

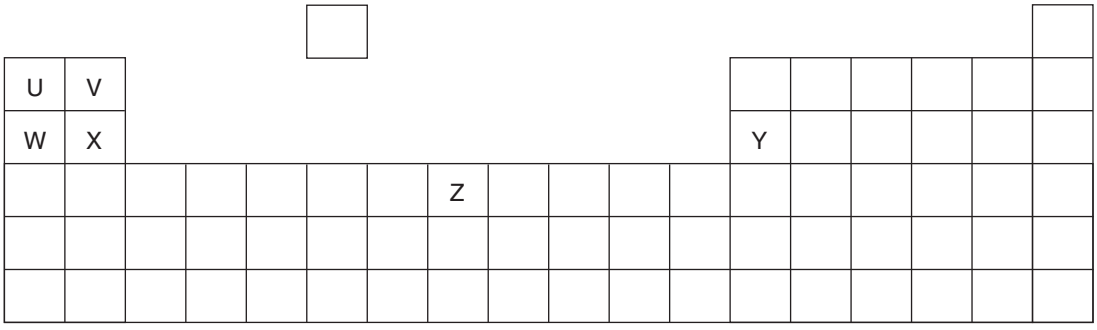


domino 骨牌

Part II Multiple choice questions

Directions: Use the following information to answer questions 7 to 9.

The positions of six elements in the periodic table are shown by the letters U, V, W, X, Y and Z (the letters are NOT symbols of the actual elements).



7 Which letter shows in the position of the alkali metal in Period 2?

(A) U Alkali metals are Group I elements. The alkali metal in Period 2 is lithium.
B V
C W
D X

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Mar. 2012, 25)

8 Which letter shows the position of the element which has chemical properties most similar to those of element X?

(A) V Elements V and X belong to the same group.
B W
C Y
D Z

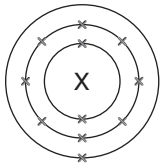
(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Mar. 2012, 26)

9 The compounds of one element produce a yellow flame in a flame test. Which letter shows the position of this element?

A U
(B) W Sodium produces a yellow flame in a flame test. Sodium is a Group I element in Period 3.
C X
D Z

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Mar. 2012, 28)

10 The diagram shows the electrons in an atom of an element X.



In the periodic table this element will be in

A Group II and Period 1.
B Group II and Period 3.
C Group I and Period 2.
(D) Group I and Period 3.

The atom has one outermost shell electron. Thus, the element belongs to Group I.
The atom has three occupied electron shells. Thus, the element is in Period 3.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Nov. 2011, 30)

11 J and Q are two alkaline earth metals in the periodic table. If the atomic number of J is x, then the atomic number of Q could be

A $x - 2$.
B $x + 6$.
C $x - 10$.
(D) $x + 18$.

(HKCEE, Paper 2, 2008, 2)

12 Which of the following statements about the periodic table is correct?

A The melting point of elements decreases across the third period.
B The melting point of Group I elements increases down the group.
C The reactivity of Group VII elements increases down the group.
(D) All elements in Group 0 exist in gaseous state at room temperature and pressure.

Option A — The melting points of elements do NOT vary gradually across a period.
Option B — The melting points of Group I elements decrease down the group.
Option C — The reactivity of Group VII elements decreases down the group.

13 The following table lists the relative reactivity of three Group I metals with water.

Metal	Relative reactivity with water
X	slow
Y	explosive
Z	vigorous

Which of the following rows shows the metals in order of increasing atomic number?

A X < Y < Z
(B) X < Z < Y
C Y < X < Z
D Y < Z < X

The reactivity of Group I elements increases down the group, i.e. increases with the atomic number.
The order of reactivity of the three metals is X < Z < Y, this is the same as the order of increasing atomic number.

14 The table shows the boiling points of four halogens. Which halogen has the lowest atomic number?

Halogen	Boiling point (°C)
A	184
(B)	–188
C	–34
D	59

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2012, 25)

15 How many electrons and neutrons are there in a doubly charged cation formed from $^{55}_{25}Mn$?

	Number of electrons	Number of neutrons
(A)	23	30
B	23	32
C	25	28
D	25	30

(HKCEE, Paper 2, 2010, 1)

14 The boiling points of halogens increase down the group i.e. increase with atomic numbers.
Halogen B has the lowest boiling point. Thus, it has the lowest atomic number.

16 Which of the following statements concerning the periodic table is / are correct?

(1) Ni is an example of transition elements.
(2) The elements are arranged in increasing order of neutron number.
(3) The lower the element located in each group, the more reactive the element is.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2009, 18)

17 Which of the following safety measures should be taken when investigating the reaction between sodium and water?

(1) Use forceps to pick sodium.
(2) Use a small piece of sodium.
(3) Use a small amount of water.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2010, 26)

18 A small piece of sodium is dropped into a trough of water containing litmus solution. Which of the following observations is / are correct?

(1) The sodium moves rapidly on the water surface.
(2) The sodium dissolves in water and the solution turns red.
(3) The sodium burns with a lilac flame.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) The solution should turn blue as sodium hydroxide makes the solution alkaline.
(3) The sodium burns with a golden yellow flame.

19 Which of the following statements concerning potassium and calcium are correct?

(1) The atomic number of calcium is greater than that of potassium.
(2) The hardness of calcium is higher than that of potassium.
(3) The reactivity of calcium is higher than that of potassium.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) The atomic number of calcium is 20 while that of potassium is 19.
(3) The reactivity of calcium is lower than that of potassium.

20 Which of the following statements about chlorine, bromine and iodine are correct?

(1) They are all coloured substances.
(2) Their reactivity decreases with increasing relative atomic mass.
(3) They all react with metals to form salts.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(2) Their reactivity decreases down the group, i.e. decreases with increasing relative atomic mass.

21 Which of the following statements concerning the conversion of a bromine atom to a bromide ion is / are correct?

(1) The atomic number of bromine increases by 1.
(2) The number of occupied electron shells in a bromine atom is the same as that in a bromide ion.
(3) The number of electrons in a bromine atom is greater than that in a bromide ion.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1) Adding an electron to a bromine atom does NOT affect its atomic number.
(3) Bromide ion (Br^-) is a singly charged negative ion. The number of electrons in a bromine atom is smaller than that in a bromide ion by 1.

Part III Structured questions

22 The table shows the numbers of protons, neutrons and electrons in different particles, A, B, C, D and E.

	A	B	C	D	E
Number of protons	11	11	11	9	9
Number of neutrons	11	12	11	10	10
Number of electrons	11	11	10	9	10

a) Which particle has the greatest mass?
B

b) Which particle has a negative charge?
E

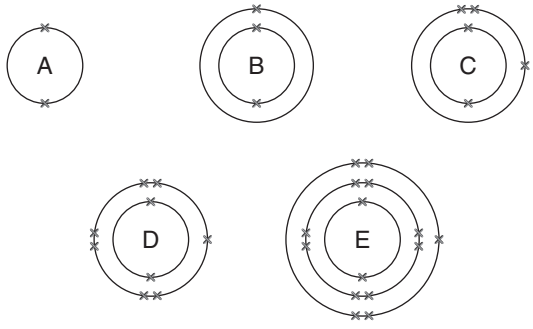
c) Which particles are atoms?
A, B, D

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 1, Jan. 2008, 5)

23 This question is about atoms.

Look at the diagrams. They show the atoms of some elements.

The letters do NOT represent the symbols for the elements.



a) i) Element B is in Group 1.

How can you tell? An atom of element B has one outermost shell electron.

ii) Element E is in Period 3.

How can you tell?

An atom of element E has three occupied electron shells.

iii) Which element forms an ion with a charge of –1?

Choose A, B, C, D or E.

Element D

b) The nucleus of an atom contains protons and neutrons.

i) Complete the table to show the relative charge and relative mass of a neutron.

	Relative charge	Relative mass
Proton	+1	1
Neutron	0	1

ii) There are two common types of chlorine atom, known as isotopes.

One isotope is called chlorine-35 and the other is called chlorine-37.

Look at the table.

It shows some information about chlorine-35.

	Symbol	Number of protons	Number of neutrons	Number of electrons
Chlorine-35	$^{35}_{17}Cl$	17	18	17
Chlorine-37	$^{37}_{17}Cl$	17	20	17

Complete the table to show information about chlorine-37.

(OCR GCSE Gateway Science, Additional Chem. B, Unit 1, Jan. 2012, 10)

24 a) Any one of the following:
- Across a period, the elements show a gradual change (increase) in non-metallic character.
- Across a period, the elements show a gradual change (decrease) in atomic size.

b) i) Their atoms have different number of occupied electron shells.

ii) Their atoms have the same number of outermost shell electrons.

c) Potassium and fluorine

d)
- Sodium floats on the surface of water while magnesium sinks.
- Sodium fizzes quickly / melts to form a silvery ball while magnesium shows no reaction.

24 A part of the periodic table is shown below:
Refer to the previous page.

Period	Group I	II	III	IV	V	VI	VII	0
2	Li	Be	B	C	N	O	F	Ne
3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca					Br	Kr
5								Xe

a) Across a period, the elements demonstrate a gradual change in some of their physical properties. State ONE such property.

b) In what way are the electronic arrangements of atoms of magnesium and calcium

i) different from each other?

ii) similar to each other?

c) Which metal and non-metal in the above table would react most vigorously with each other?

d) Sodium and magnesium are added separately to cold water in troughs. State TWO differences in the observations you expect.

25 Rubidium (Rb) and potassium (K) are elements in Group I of the periodic table.

a) What is the name commonly given to this group of elements?
Alkali metals

b) The atomic number of rubidium is 37. The electronic arrangement of a rubidium atom is $p,q,r,8,1$.

i) What are the values of p, q and r?
$p = 2$, $q = 8$, $r = 18$

ii) How does a rubidium atom change into a rubidium ion?
It does so by losing one electron.

c) Explain, in terms of electronic arrangements of their atoms, why rubidium and potassium have similar chemical properties. Their atoms have the same number of outermost shell electrons.

d) Predict, with ONE reason, whether rubidium or potassium is more reactive towards chlorine.
Rubidium is more reactive towards chlorine.
The reactivity of Group I elements increases down the group.

e) A sample of rubidium consists of two isotopes, rubidium-85 and rubidium-87. Complete the following table about the atomic structures of these two isotopes.

Isotope	rubidium-85	rubidium-87
Atomic number	37	37
Mass number	85	87
Number of protons	37	37
Number of neutrons	48	50
Number of electrons	37	37

26 The table gives information about lithium and sodium, two metals in Group I of the periodic table.

Metal	**Hardness**	**Melting point (°C)**	**Boiling point (°C)**	**Conductor of electricity**
Lithium	soft	181	1 347	good
Sodium	soft	98	883	good

a) Potassium is another metal in group I.

Predict whether you could cut a small sample of potassium with a sharp knife. Explain your answer.
Yes, potassium is soft.

b) Using only the information from the table, state

i) one property of sodium that is typical of all metals.
Good conductor of electricity

ii) one property of sodium that is NOT typical of most other metals.
Low melting point / low boiling point / soft

c) When sodium is added to water, a reaction occurs. Describe what you would see in this reaction.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C3, May 2010, 3)

26 c)
- Melts to form a silvery ball.
- Fizzes quickly.
- Moves rapidly on the water surface.

27 Sodium, potassium and lithium are Group I metals.

The Group I metals react when put into water.

Look at the table.

Metal	Time for 0.5 g of metal to react (s)	Observations
Sodium	12	• melts • skates across surface of water • gas given off • alkaline solution made
Potassium	6	• melts and catches fire • skates across surface of water • gas given off • alkaline solution made
Lithium	20	• skates across surface of water • gas given off • alkaline solution made

a) All three reactions give off the same gas.

Write down the name of this gas.
Hydrogen

b) Look at the observations for sodium.

Write down the name of the substance that makes the solution alkaline.
Sodium hydroxide

c) What is the flame colour when potassium burns?
Lilac

d) Rubidium is another element in Group I.

It is below lithium, sodium and potassium.

Predict the observations you should get when 0.5 g of rubidium reacts with water.

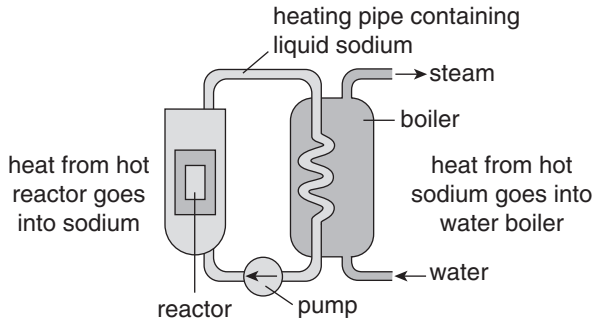
Your answer should include

- an estimate of the time it takes to react; and
- what you would see.

(OCR GCSE Gateway Science, Chem. B, Unit 1, Jan. 2011, 11)

27 d) Reaction time less than 6 seconds.
Any two of the following:
- explosive reaction (more reactive than potassium)
- skates across surface of water
- catches fire
- gas given off
- alkaline solution formed
- melts

28 In some nuclear power stations, sodium is used to carry the heat from the reactor to the boiler. Heat from the hot sodium turns the water in the boiler into steam.



The sodium must be melted so that it can flow through the pipes.

Sodium is a Group I metal.

Here is some information about Group I metals.

	Melting point (°C)	Boiling point (°C)
Lithium	180 °C	1 342 °C
Sodium		883 °C
Potassium	63 °C	760 °C

a) Estimate the melting point of sodium.
88–145 °C

b) A student thinks that using sodium in a nuclear power station might cause problems.

He has two reasons:

- the melting point makes it difficult to use sodium in pipes; and
- the sodium might be dangerous if the pipes leak inside the boiler.

Suggest and explain what these problems might be.

c) When sodium reacts it makes a positive ion.
Loses electron(s).
What happens when an atom changes into a positive ion?

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit A151/02, May 2012, 4)

28 b) Sodium needs to be a liquid / needs to be kept hot.
Sodium is very reactive / sodium reacts with water.
Hydrogen is produced and the gas may lead to explosion.

Unit Exercise

29 a) The table shows the electronic arrangements of atoms of the elements in Period 3 of the periodic table.

Element	Electronic arrangement
Na	2,8,1
Mg	2,8,2
Al	2,8,3
Si	2,8,4
P	2,8,5
S	2,8,6
Cl	2,8,7
Ar	2,8,8

i) How is the electronic arrangement of an atom of an element related to its position in the periodic table?

ii) Give the electronic arrangement of an atom of the element directly below magnesium in the periodic table.

b) Explain the meaning of the term 'isotopes'.

c) An element exists as three isotopes. The table gives some information about them.

Number of neutrons	12	13	14
Number of protons	12	12	12
Atomic number of isotope	12	12	12
Mass number of isotope	24	25	26
Percentage of each isotope in the element	79.0	10.0	11.0

i) Complete the table for the isotopes of the element.

ii) Use the information in the table to identify the element.

iii) Use the information in the table to calculate the relative atomic mass of the element. 24.3

iv) When a sample of the element containing only atoms with a mass number of 24 was added to dilute sulphuric acid, effervescence was seen.

What would be seen if a sample of the element containing only atoms with a mass number of 26 was added to dilute sulphuric acid? Explain your answer.

(Edexcel IGCSE, Chemistry, Paper 2, Nov. 2008, 5)

30 Both bromine and chlorine are Group VII elements in the periodic table.

a) What is the name commonly given to this group of elements?
Halogens

b) Draw an electron diagram of a chlorine atom.

c) The electronic arrangement of a bromine atom is 2,8, *p*, *q*.

i) What are the values of *p* and *q*?
p = 18, *q* = 7

ii) To which period of the periodic table does bromine belong? Explain your answer. Period 4
Bromine has 4 occupied electron shells in its atom.

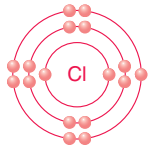
iii) How does a bromine atom change into a bromide ion?
It does so by gaining one electron.

iv) In terms of protons, neutrons and electrons, how is a bromine atom similar to a bromide ion.
A bromine atom and a bromide ion have the same number of protons and neutrons.

d) Explain, in terms of electronic arrangements of their atoms, why chlorine and bromine have similar chemical properties. Their atoms have the same number of outermost shell electrons.

e) Suggest whether the reactivity of Group VII elements increases or decreases down the group.
Decreases

30 b)



29 a) i) Number of outermost shell electrons in an atom of an element in the same as the group number of element.
Number of occupied electron shells in an atom of an element is the same as the period number of the element.
ii) 2,8,8,2
b) Isotopes are different atoms of an element which have the same number of protons but a different number of neutrons.

c) ii) Magnesium
iii) Relative atomic mass of magnesium
$= \frac{24 \times 79.0 + 25 \times 10.0 + 26 \times 11.0}{100}$
= 24.3
iv) Observation: The same
Explanation: The atoms have the same number of electrons.

31 The noble gases are in Group 0 of the periodic table.

a) Argon is the most abundant noble gas.

It has been used for many years in light bulbs.

i) The atomic number of argon is 18.

Give the electronic arrangement of an argon atom.

2,8,8

ii) State and explain, in terms of the electronic arrangement, why argon is used in light bulbs.

Argon is unreactive because its atom has 8 outermost shell electrons.

b) Neon exists as two isotopes.

Isotope	Atomic number	Mass number	Relative abundance (%)
Neon-20	10	20	90.9
Neon-22	10	22	9.10

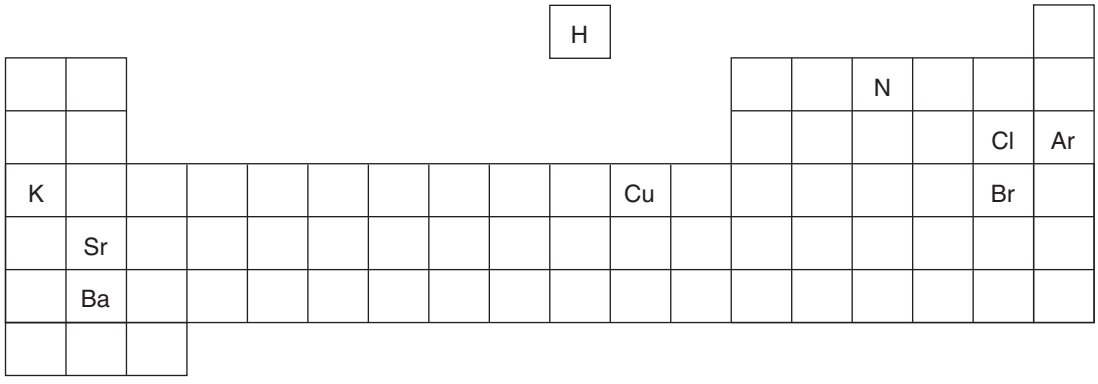
Calculate the relative atomic mass of neon.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2009, 3)

Relative atomic mass of neon = $\frac{20 \times 90.9 + 22 \times 9.10}{100}$

= 20.2

32 Look at the diagram. It shows an outline of the periodic table.



Choose your answer ONLY from the symbols shown on the outline table.

Use the complete periodic table to help you.

Which symbol shows

a) an element whose atoms have 8 electrons in their outermost shell?

Ar

b) an element that is a red / brown liquid?

Br

c) an element that forms blue compounds?

Cu

d) the element in Group 2 and Period 5?

Sr

(OCR GCSE Gateway Science, Additional Science B, Unit 1, Jan. 2010, 22)

Unit Exercise